PORTFOLIO SAMPLE · RESEARCH PAPER SUMMARY

Dexamethasone in Hospitalised COVID-19: The RECOVERY Trial

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Medical Research Summary · Oct 2026

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6,425
patients randomised (2,104 vs 4,321)

176
NHS hospital organisations, UK

6 mg/day
dexamethasone, up to 10 days

0.83
rate ratio for 28-day death (95% CI 0.75 to 0.93)

1. Research question

In patients hospitalised with COVID-19, does a low dose of dexamethasone, a cheap and widely available corticosteroid, reduce death at 28 days compared with usual care alone?

2. Methods

- **Design:** randomised, controlled, open-label, adaptive platform trial (one arm of the larger RECOVERY platform).
- **Population:** hospitalised patients with suspected or confirmed SARS-CoV-2 infection, recruited March to June 2020.
- **Intervention:** dexamethasone 6 mg once daily, oral or intravenous, for up to 10 days or until discharge. Median treatment 7 days.
- **Comparator:** usual standard of care, with 2:1 allocation to usual care.
- **Primary outcome:** all-cause mortality within 28 days of randomisation, analysed by intention to treat with age adjustment.

3. Results

Overall, 22.9% of patients given dexamethasone died within 28 days, compared with 25.7% on usual care (rate ratio 0.83; 95% CI 0.75 to 0.93; P<0.001). The effect depended strongly on how much breathing support patients needed when they entered the trial:

Respiratory support at randomisation	Dexamethasone	Usual care	Rate ratio (95% CI)
Invasive mechanical ventilation	29.3%	41.4%	0.64 (0.51 to 0.81)
Oxygen only	23.3%	26.2%	0.82 (0.72 to 0.94)
No oxygen	17.8%	14.0%	1.19 (0.91 to 1.55)

Benefit was largest in ventilated patients, where roughly one death was prevented for every eight patients treated. There was no evidence of benefit in patients not receiving oxygen, and the point estimate leaned towards harm.

4. Critical appraisal

STRENGTHS

- Large sample and hard primary outcome (death).
- Pragmatic design across many hospitals, so results apply to real-world care.
- Pre-specified subgroups by respiratory support explain where the benefit lies.
- Cheap, available drug: rapid global uptake possible.

LIMITATIONS

- Open-label, though mortality is unlikely to be biased by awareness of treatment.
- Limited data on adverse effects such as hyperglycaemia and secondary infections.
- Conducted in UK hospitals early in the pandemic, before vaccines and newer variants.
- Subgroup findings, while pre-specified, carry less certainty than the overall result.

5. What this means for practice

Dexamethasone became standard care for hospitalised COVID-19 patients who need supplemental oxygen or mechanical ventilation, and international guidance adopted this soon after the results were released. It should not be given routinely to patients who do not need oxygen.

6. Plain-language summary

Researchers in the UK tested whether a cheap steroid tablet, dexamethasone, could save the lives of people in hospital with COVID-19. Among more than 6,000 patients, those given the drug were less likely to die within four weeks. The benefit was greatest for the sickest patients, those on a breathing machine, and smaller for those on oxygen. People who did not need oxygen did not benefit, so the drug is only recommended for patients who need help with their breathing.

Independent summary prepared as a portfolio sample. Figures are taken from the published paper; readers should consult the original article for full details. Not medical advice for any individual.